markers of apoptosis and autophagy in cells exposed to CBD and THC with or without irradiation for 4 hours, as these have been previously implicated in cannabinoid action. Generally, our results showed that irradiation enhanced the actions of the cannabinoids. Specifically, the cannabinoids, when used at higher concentrations, decreased pAKT and pERK levels, and these were reduced further when combined with irradiation. Similarly, cannabinoids induced autophagy in the cells as indicated by a dose-dependent increase in the ratio of LC3Bii:LC3Bi, which again was further enhanced in the irradiation combination group. This pattern was not replicated by the markers of apoptosis; however, the only time that cleavage of caspase-3 was seen was when the higher dose of cannabinoids was combined with irradiation (Fig. 3D).

CBD, THC, and radiotherapy is a superior treatment regimen in mice

The efficacy of a treatment regimen consisting of CBD, THC, and radiotherapy was subsequently assessed in a murine model. The growth of the GL261 glioma tumor orthotopically implanted in C57BL/6 mice was tracked by MRI technology, and the effects on it of either pure CBD and THC together, irradiation, or the combination of both treatment modalities were determined (Fig. 4A). Of the mice inoculated with tumor cells, 80% developed tumors. Tumors in control mice were visible around day 9 and reached their permissible volumes by day 21, at which time all animals were sacrificed. Animals were exposed to suboptimal doses of each of the treatments, which allowed a window through which any improved combinatorial activity could be seen. Results

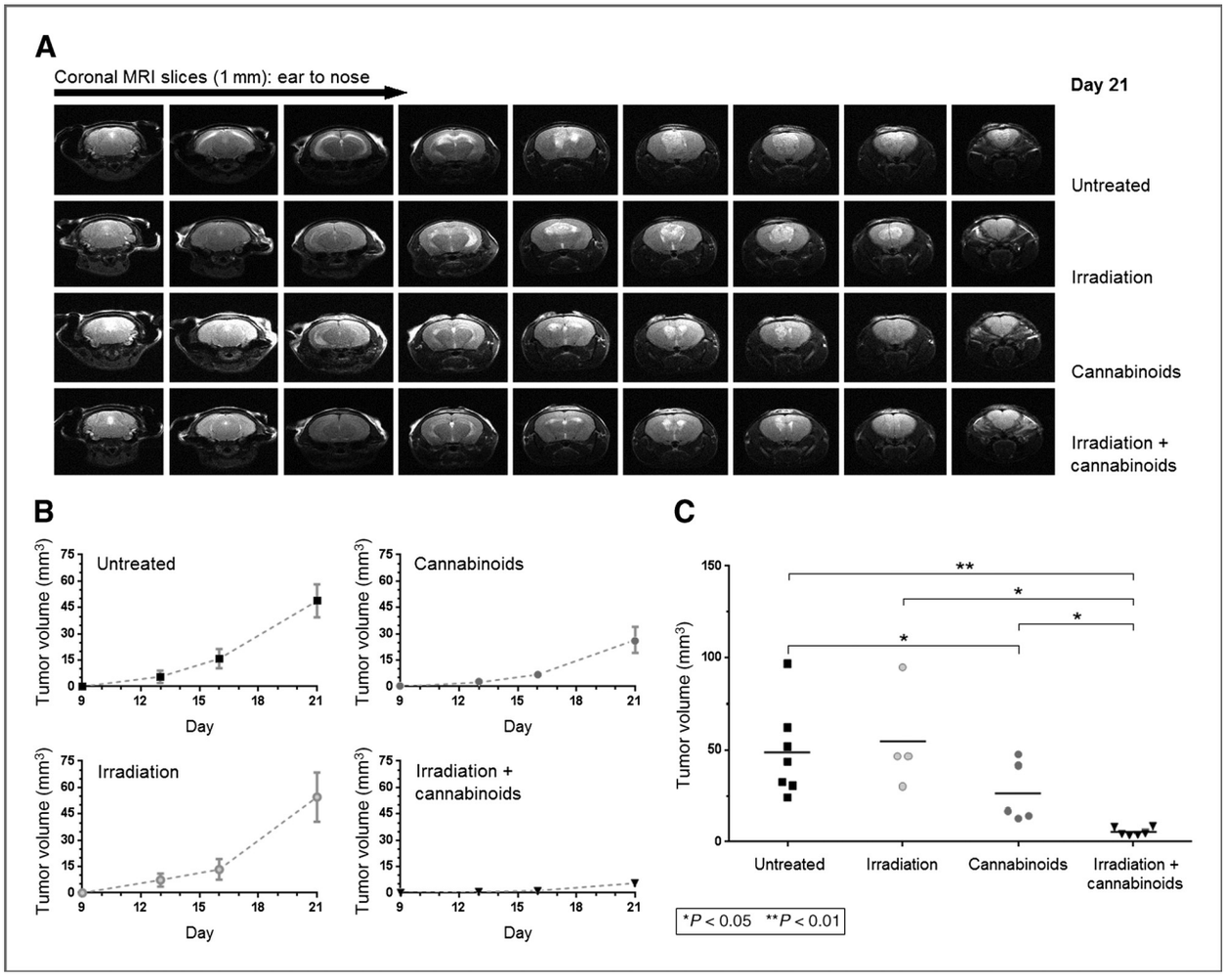

Figure 4. Effect of CBD+THC and irradiation in an orthotopic murine model. GL261 cells were injected intracranially into C57BL/6 mice, and the effect on tumor growth of pure CBD and pure THC together, irradiation, or a combination of the two modalities was assessed. MRI scans were performed on days 9, 13, 16, and 21. Representative images from MRI are shown in A, which shows the presence of tumor masses within the brain. The rate of tumor growth was much slower in the group of mice treated with CBD, THC, and irradiation (B). Furthermore, the final tumor sizes were significantly smaller in this group compared with the other three groups (C). *P* values were established from paired *t* tests following determinations of differences between variable and control groups using one-way ANOVA.

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